

MVLU COLLEGE

TABLEAU PRACTICAL NO. 9

AIM: Cleaning and Structuring Data for Analysis

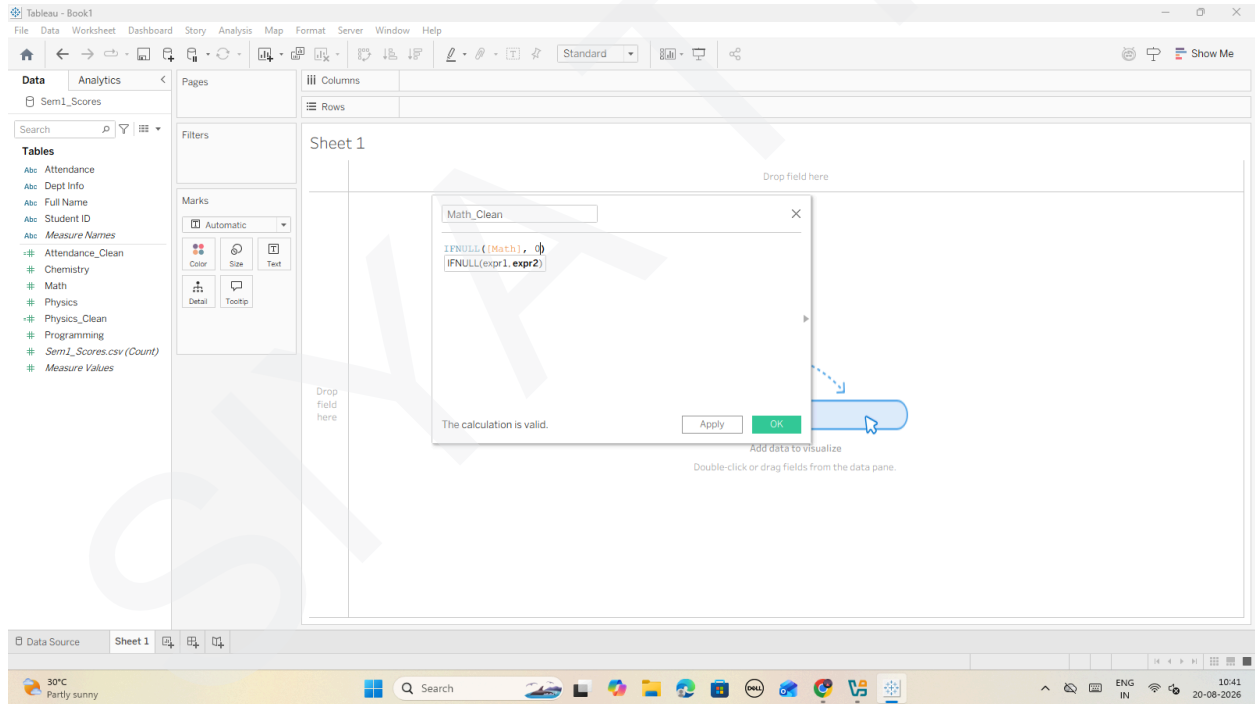
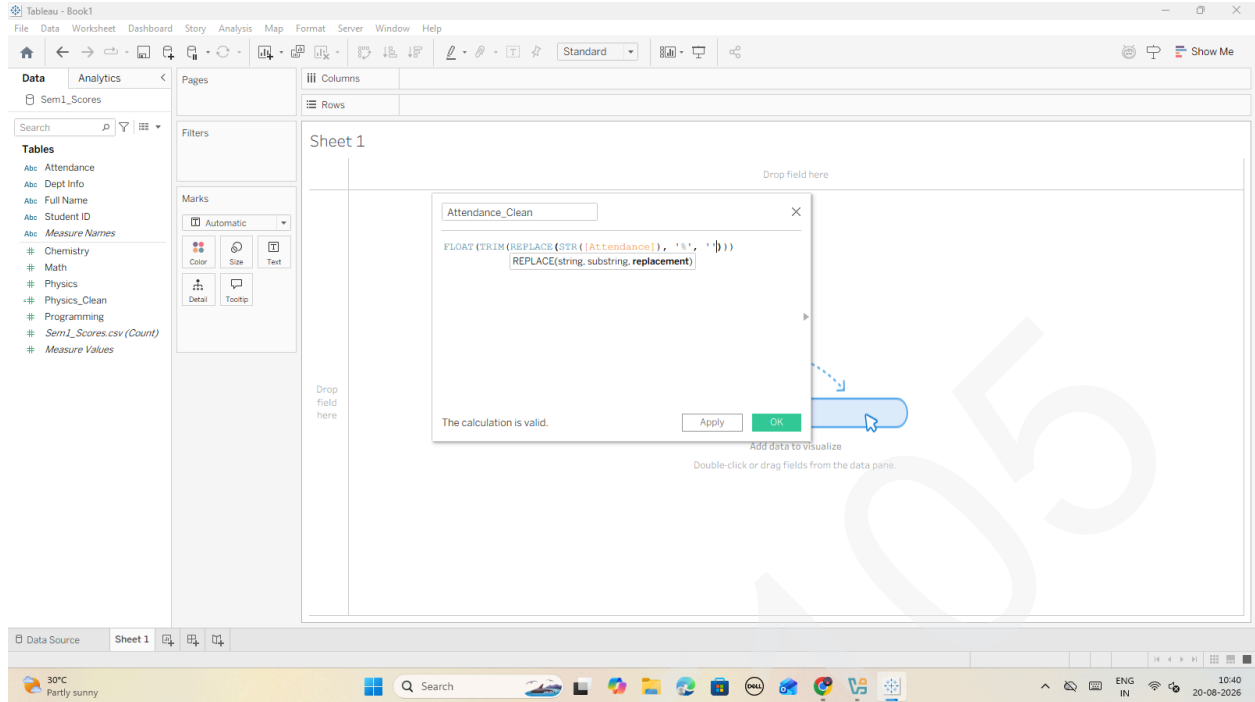
- Identify and handle missing or inconsistent data
- Convert wide data into tall format

The screenshot displays the Tableau Desktop interface. The top pane shows the 'Sem1_Scores' data source, which is a text file with 9 fields and 10 rows. The fields are: Student ID, Full Name, Dept Info, Math, Physics, Chemistry, Programming, and Attendance. The bottom pane shows the 'Columns' and 'Rows' shelves, which are currently empty. A 'Marks' card is also present, set to 'Automatic'. A dialog box titled 'Physics_Clean' is open, showing a calculated field formula: `Float(Replace(Str([Physics]), ' ', ''))`. The dialog box includes an 'Apply' button and an 'OK' button. The background of the interface is watermarked with 'SIYA POIPKAR T105'.

Student ID	Full Name	Dept Info	Math	Physics	Chemistry	Programming	Attendance
S101	Aarav Sharma	CSE-01	85	0.900000	88	95	85 %
S102	priya VERMA	ECE-02	null	0.750000	80	85	90 %
S103	Rohan Gupta	CSE-01	78	0.820000	null	88	78 %
S104	Sneha Patel	IT-03	92	0.880000	90	96	95 %
S105	aditya Rao	MECH-04	45	null	50	60	65 %
S106	Ananya Singh	CSE-01	88	0.910000	85	92	88 %
S107	Rahul Nair	ECE-02	null	0.700000	75	80	72 %

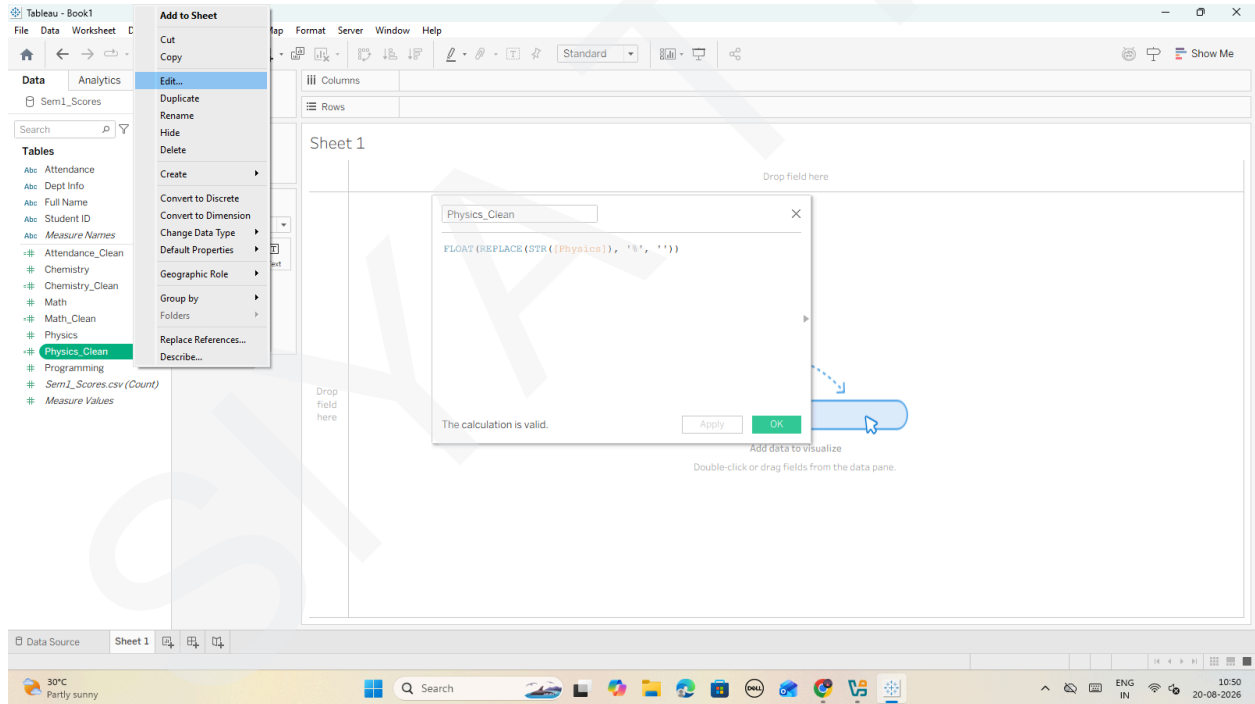
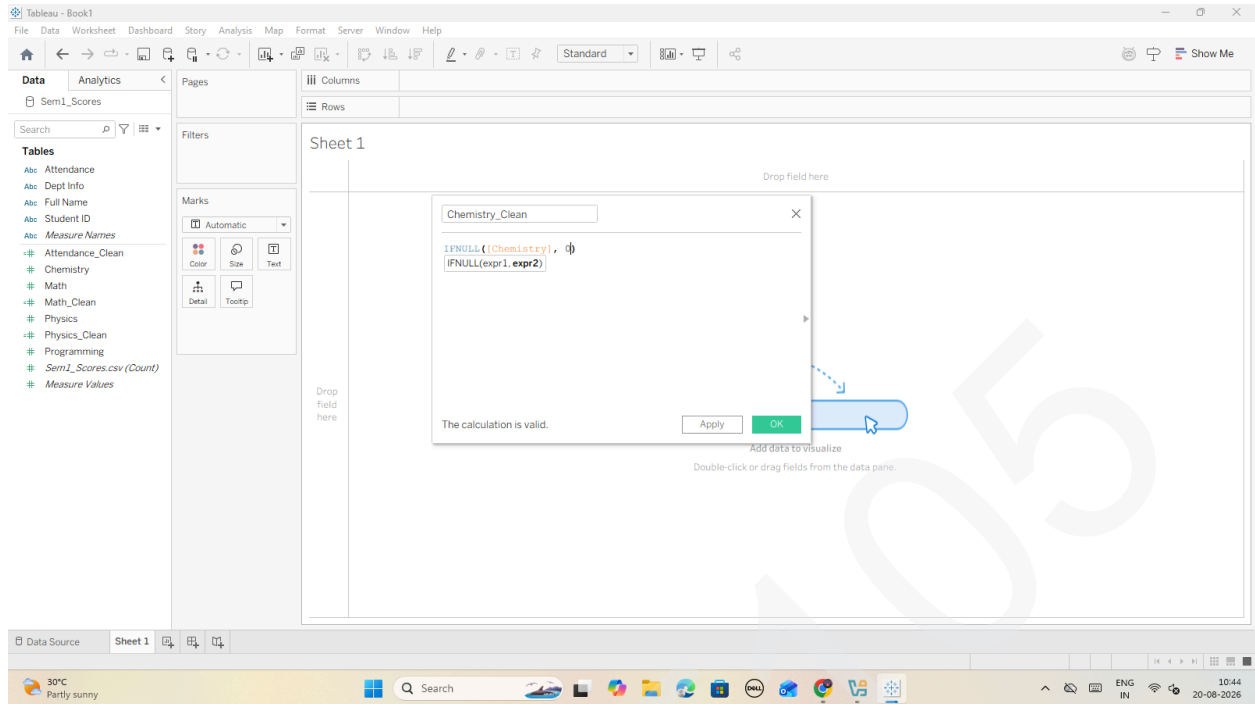
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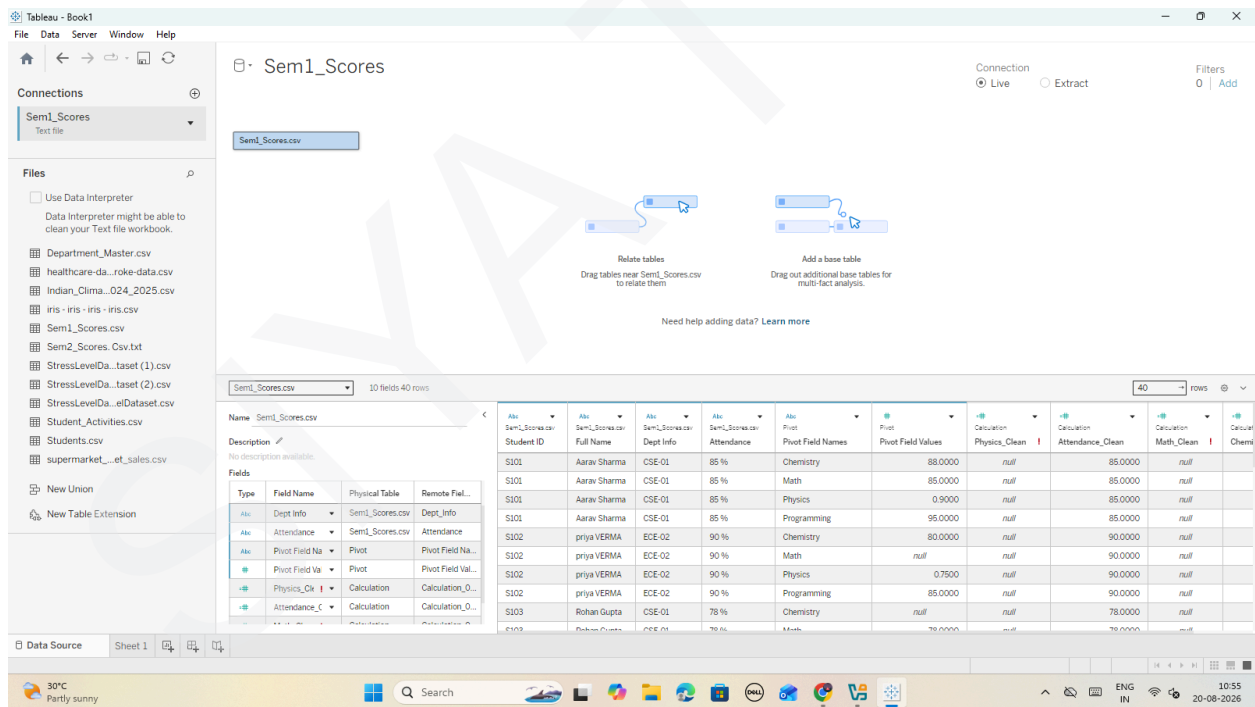
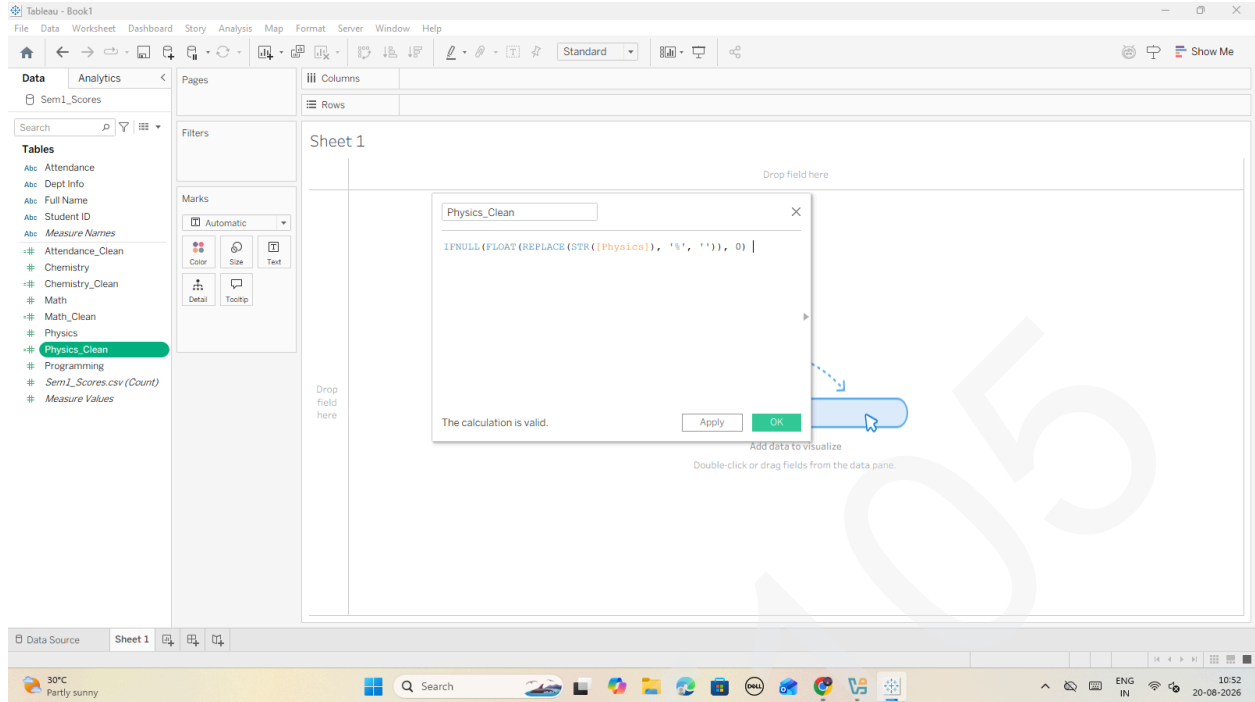
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TABLEAU PRACTICAL NO. 9



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TABLEAU PRACTICAL NO. 9



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TABLEAU PRACTICAL NO. 9

Tableau - Book1

File Data Server Window Help

Connections

Sem1_Scores
Text file

Files

Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

Department_Master.csv
healthcare-da_roke-data.csv
Indian_Clima_024_2025.csv
iris - iris - iris.csv
Sem1_Scores.csv
Sem2_Scores.Csv.txt
StressLevelDa_taset (1).csv
StressLevelDa_taset (2).csv
StressLevelDa_eDataset.csv
Student_Activities.csv
Students.csv
supermarket_et_sales.csv

New Union
New Table Extension

Sem1_Scores

10 fields 40 rows

Name Sem1_Scores.csv

Description No description available.

Fields

Type	Field Name	Physical Table	Remote File...
Alt	Attendance	Sem1_Scores.csv	Attendance
Alt	Subject	Pivot	Pivot Field Na...
+	Score	Pivot	Pivot Field Val...
+	Physics_Ck	Calculation	Calculation_O...
+	Attendance_C	Calculation	Calculation_O...
+	Math_Clear	Calculation	Calculation_O...

40 rows

Sem1_Scores.csv Student ID	Sem1_Scores.csv Full Name	Sem1_Scores.csv Dept Info	Sem1_Scores.csv Attendance	Alt Pivot Subject	+ <th>+ <th>+ <th>+ <th>+ </th></th></th></th>	+ <th>+ <th>+ <th>+ </th></th></th>	+ <th>+ <th>+ </th></th>	+ <th>+ </th>	+
S101	Aarav Sharma	CSE-01	85 %	Chemistry	88.0000	Calculation Physics_Clean	Calculation Attendance_Clean	Calculation Math_Clean	Calculation Chem...
S101	Aarav Sharma	CSE-01	85 %	Math	85.0000	null	85.0000	null	
S101	Aarav Sharma	CSE-01	85 %	Physics	0.9000	null	85.0000	null	
S101	Aarav Sharma	CSE-01	85 %	Programming	95.0000	null	85.0000	null	
S102	priya VERMA	ECE-02	90 %	Chemistry	80.0000	null	90.0000	null	
S102	priya VERMA	ECE-02	90 %	Math	null	null	90.0000	null	
S102	priya VERMA	ECE-02	90 %	Physics	0.7500	null	90.0000	null	
S102	priya VERMA	ECE-02	90 %	Programming	85.0000	null	90.0000	null	
S103	Rohan Gupta	CSE-01	78 %	Chemistry	null	null	78.0000	null	
S103	Rohan Gupta	CSE-01	78 %	Math	78.0000	null	78.0000	null	

30°C Partly sunny

Search

ENG IN 10:56 20-08-2026

Tableau - Book1

File Data Server Window Help

Connections

Sem1_Scores
Text file

Files

Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

Department_Master.csv
healthcare-da_roke-data.csv
Indian_Clima_024_2025.csv
iris - iris - iris.csv
Sem1_Scores.csv
Sem2_Scores.Csv.txt
StressLevelDa_taset (1).csv
StressLevelDa_taset (2).csv
StressLevelDa_eDataset.csv
Student_Activities.csv
Students.csv
supermarket_et_sales.csv

New Union
New Table Extension

Sem1_Scores

10 fields 40 rows

Name Sem1_Scores.csv

Description No description available.

Fields

Type	Field Name	Physical Table	Remote File...
Alt	Attendance	Sem1_Scores.csv	Attendance
Alt	Subject	Pivot	Pivot Field Na...
+	Score	Pivot	Pivot Field Val...
+	Physics_Ck	Calculation	Calculation_O...
+	Attendance_C	Calculation	Calculation_O...
+	Math_Clear	Calculation	Calculation_O...

40 rows

Sem1_Scores.csv Student ID	Sem1_Scores.csv Full Name	Sem1_Scores.csv Dept Info	Sem1_Scores.csv Attendance	Alt Pivot Subject	+ <th>+ <th>+ <th>+ <th>+ </th></th></th></th>	+ <th>+ <th>+ <th>+ </th></th></th>	+ <th>+ <th>+ </th></th>	+ <th>+ </th>	+
S101	Aarav Sharma	CSE-01	85 %	Chemistry	88.0000	Calculation Physics_Clean	Calculation Attendance_Clean	Calculation Math_Clean	Calculation Chem...
S101	Aarav Sharma	CSE-01	85 %	Math	85.0000	null	85.0000	null	
S101	Aarav Sharma	CSE-01	85 %	Physics	0.9000	null	85.0000	null	
S101	Aarav Sharma	CSE-01	85 %	Programming	95.0000	null	85.0000	null	
S102	priya VERMA	ECE-02	90 %	Chemistry	80.0000	null	90.0000	null	
S102	priya VERMA	ECE-02	90 %	Math	null	null	90.0000	null	
S102	priya VERMA	ECE-02	90 %	Physics	0.7500	null	90.0000	null	
S102	priya VERMA	ECE-02	90 %	Programming	85.0000	null	90.0000	null	
S103	Rohan Gupta	CSE-01	78 %	Chemistry	null	null	78.0000	null	
S103	Rohan Gupta	CSE-01	78 %	Math	78.0000	null	78.0000	null	

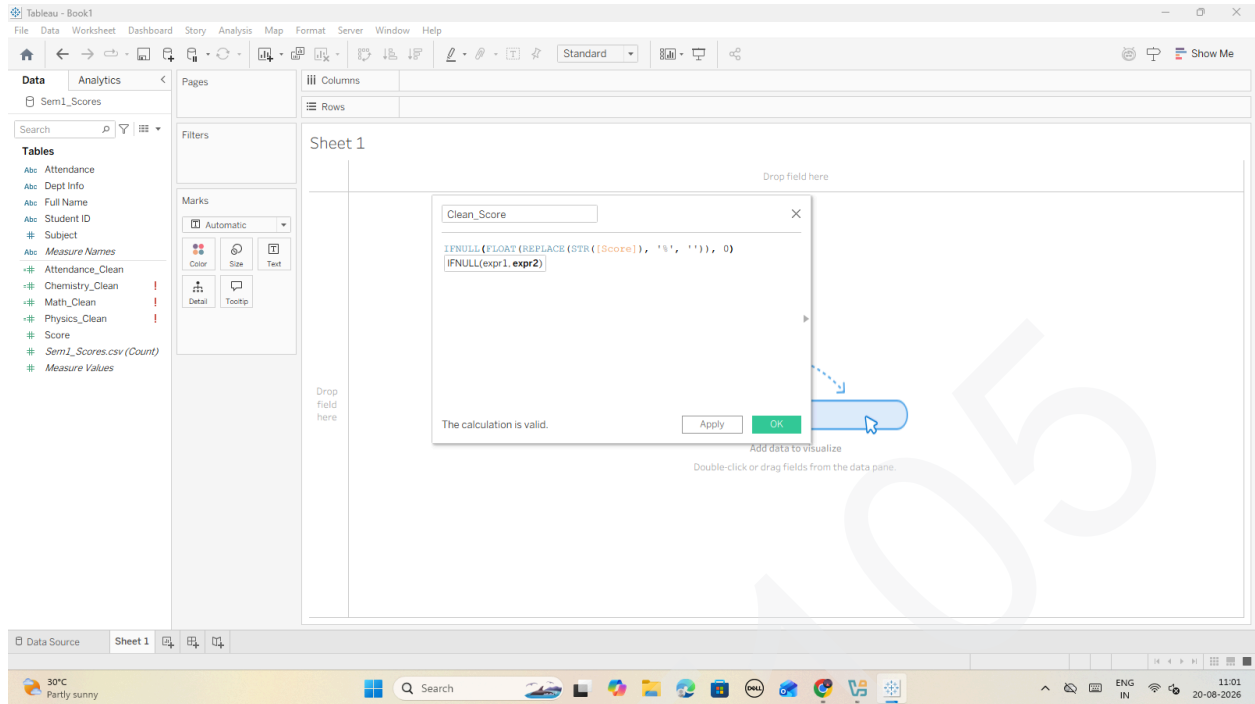
30°C Partly sunny

Search

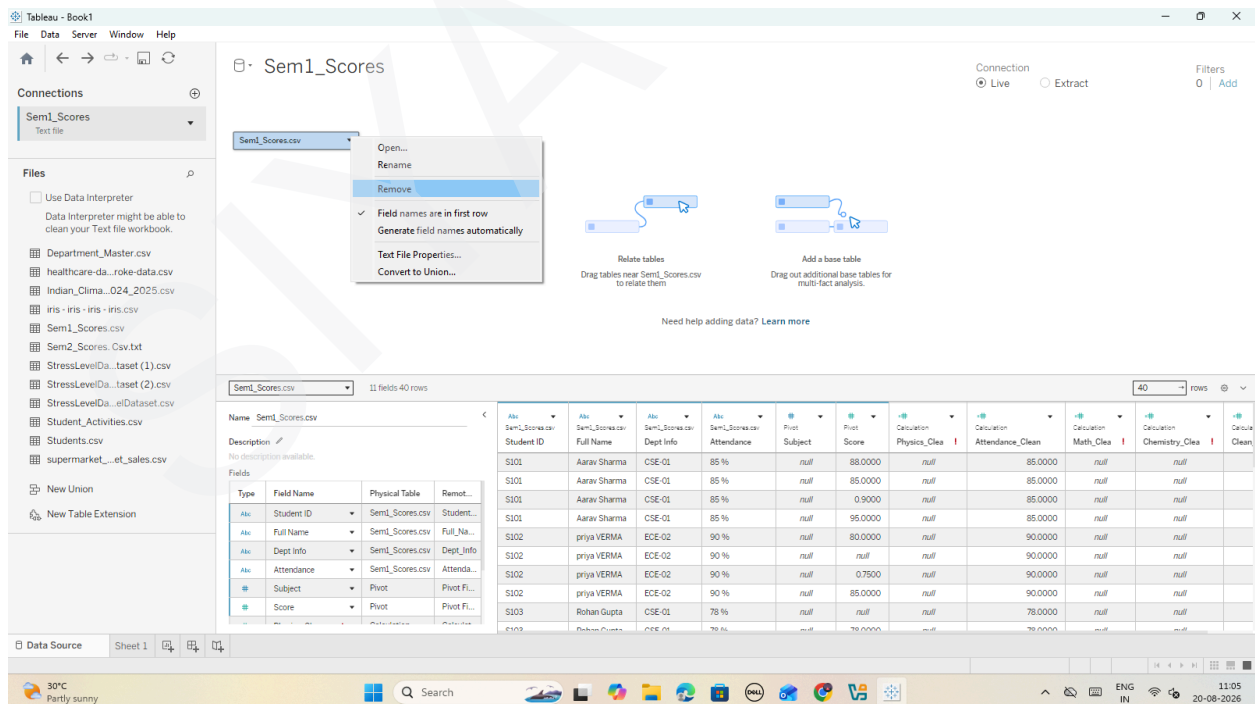
ENG IN 10:57 20-08-2026

MVLU COLLEGE

TABLEAU PRACTICAL NO. 9

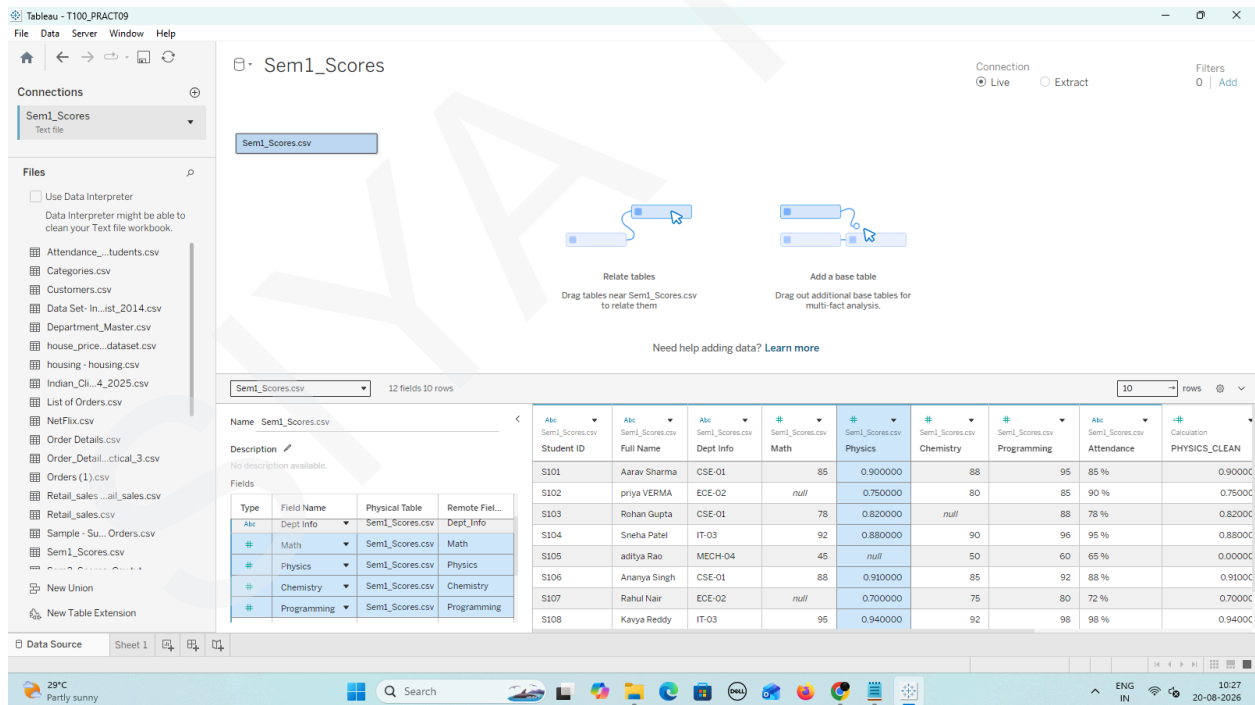
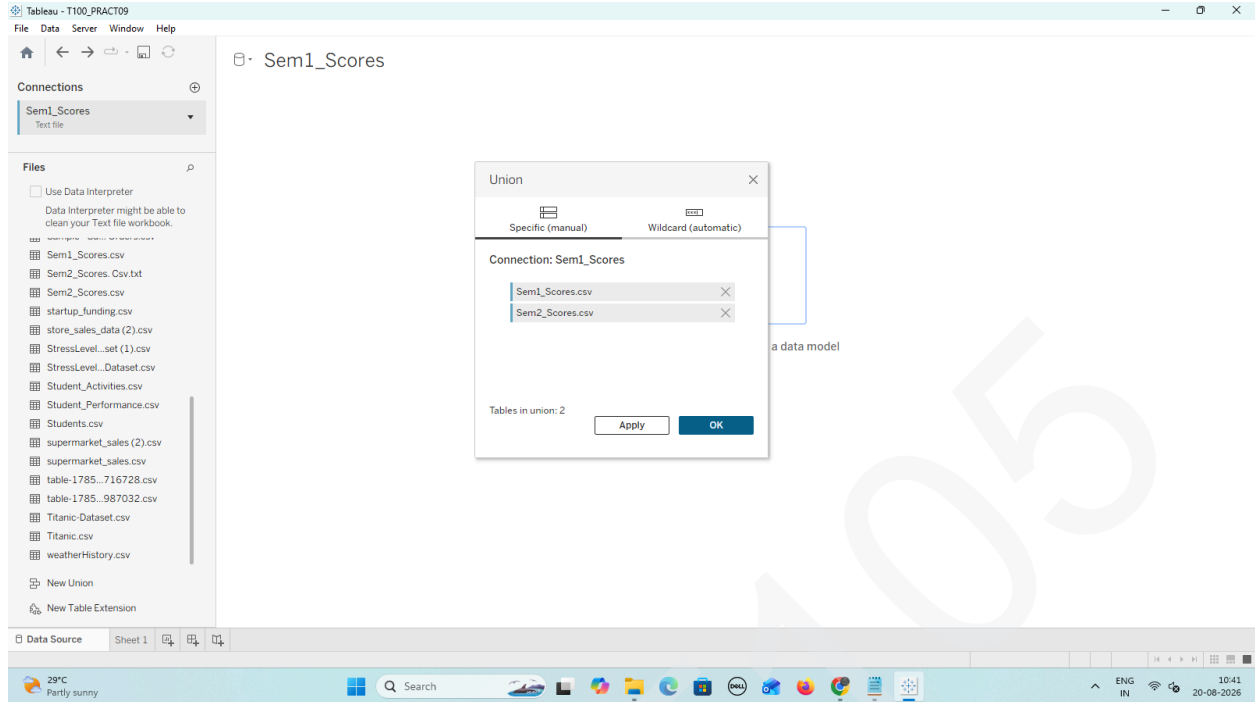


- c) Perform union of multiple files
- d) Execute cross-database joins



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TABLEAU PRACTICAL NO. 9



File

Data

Server

Window

Help

Home

Back

Forward

Refresh

Download

Connections

Sem1_Scores

Text file

Files

Use Data Interpreter

Data Interpreter might be able to clean your Text file workbook.

Attendance...tudents.csv

Categories.csv

Customers.csv

Data Set - In...ist_2014.csv

Department_Master.csv

house_price...dataset.csv

housing - housing.csv

Indian_Cli...4_2025.csv

List of Orders.csv

NetFix.csv

Order Details.csv

Order_Detail...ctical_3.csv

Orders (1).csv

Retail_sales...all_sales.csv

Retail_sales.csv

Sample - Sou...Orders.csv

Sem1_Scores.csv

Sem1_Scores.csv

New Union

New Table Extension

Sem1_Scores

Relate tables

Drag tables near Sem1_Scores.csv to relate them

Add a base table

Drag out additional base tables for multi-fact analysis.

Need help adding data? [Learn more](#)

Sem1_Scores.csv

12 fields 10 rows

10 rows

Name

Sem1_Scores.csv

Description

No description available.

Fields

Type

Field Name

AK1

Dept Info

#

Math

#

Physics

#

Chemistry

#

Programming

Rename

Copy Values

Hide

Create Calculated Field...

Pivot

Merge Mismatched Fields

Sem1_Scores.csv

Chemistry

Sem1_Scores.csv

Programming

AK1	AK1	AK1	Sem1_Scores.csv	Sem1_Scores.csv	Sem1_Scores.csv	Sem1_Scores.csv	Sem1_Scores.csv	Sem1_Scores.csv	Calculation
Student ID	Full Name	Dept Info	Math	Physics	Chemistry	Programming	Attendance	PHYSICS_CLEAN	
S101	Aarav Sharma	CSE-01	85	0.900000	88	95	85%	0.900000	
S102	priya VERMA	ECE-02	null	0.750000	80	85	90%	0.750000	
S103	Rohan Gupta	CSE-01	78	0.820000	null	88	78%	0.820000	
S104	Sneha Patel	IT-03	92	0.880000	90	96	95%	0.880000	
S105	aditya Rao	MECH-04	45	null	50	60	65%	0.000000	
S106	Ananya Singh	CSE-01	88	0.910000	85	92	88%	0.910000	
S107	Rahul Nair	ECE-02	null	0.700000	75	80	72%	0.700000	
S108	Kavya Reddy	IT-03	95	0.940000	92	98	98%	0.940000	

Data Source

Sheet 1

Table

Grid

Download

Print

29°C

Partly sunny

Windows

Search

Taskbar

10:27

20-08-2024

SIYA POIPKAR T105

MVLU COLLEGE

TABLEAU PRACTICAL NO. 9

Tableau - T100_PRACT09

File Data Server Window Help

Connections

Sem1_Scores
Text file

Files

Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

Sem1_Scores.csv
Sem2_Scores.Csv.txt
Sem2_Scores.csv
startup_funding.csv
store_sales_data (2).csv
StressLevel...set (1).csv
StressLevel...Dataset.csv
Student_Activities.csv
Students.csv
supermarket_sales (2).csv
supermarket_sales.csv
table-1785...716728.csv
table-1785...987032.csv
Titanic-Dataset.csv
Titanic.csv
weatherHistory.csv

New Union
New Table Extension

Sem1_Scores

Relate tables
Drag tables near Sem1_Scores.csv to relate them

Add a base table
Drag out additional base tables for multi-fact analysis.

Need help adding data? [Learn more](#)

Sem1_Scores.csv 10 fields 40 rows

Name Sem1_Scores.csv

Description
No description available.

Fields

Type	Field Name	Physical Table	Remote...
Alt	Attendance	Sem1_Scores.csv	Attendan...
Alt	Subject	Pivot	Pivot Fiel...
+	Score	Pivot	Pivot Fiel...
+	PHYSICS_CLEAN	Calculation	Calculati...
+	Attendance_clean	Calculation	Calculati...

Sem1_Scores.csv	Sem1_Scores.csv	Sem1_Scores.csv	Sem1_Scores.csv	Sem1_Scores.csv	Pivot	Score	Calculation	PHYSICS_CLEAN	Calculation	Attendance_clean
Student ID	Full Name	Dept info	Attendance	Subject						
S101	Aarav Sharma	CSE-01	85 %	Chemistry		88.0000		null		85.00
S101	Aarav Sharma	CSE-01	85 %	Math		85.0000		null		85.00
S101	Aarav Sharma	CSE-01	85 %	Physics		0.9000		null		85.00
S101	Aarav Sharma	CSE-01	85 %	Programming		95.0000		null		85.00
S102	priya VERMA	ECE-02	90 %	Chemistry		80.0000		null		90.00
S102	priya VERMA	ECE-02	90 %	Math		null		null		90.00
S102	priya VERMA	ECE-02	90 %	Physics		0.7500		null		90.00
S102	priya VERMA	ECE-02	90 %	Programming		85.0000		null		90.00

30°C Partly sunny

Search

ENG IN 10:55 20-08-2026

Tableau - T100_PRACT09

File Data Server Window Help

Connections

Sem1_Scores
Text file

Files

Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

Sem1_Scores.csv
Sem2_Scores.Csv.txt
Sem2_Scores.csv
startup_funding.csv
store_sales_data (2).csv
StressLevel...set (1).csv
StressLevel...Dataset.csv
Student_Activities.csv
Students.csv
supermarket_sales (2).csv
supermarket_sales.csv
table-1785...716728.csv
table-1785...987032.csv
Titanic-Dataset.csv
Titanic.csv
weatherHistory.csv

New Union
New Table Extension

Sem1_Scores

Relate tables
Drag tables near Sem1_Scores.csv to relate them

Add a base table
Drag out additional base tables for multi-fact analysis.

Need help adding data? [Learn more](#)

Sem1_Scores.csv 10 fields 40 rows

Name Sem1_Scores.csv

Description
No description available.

Fields

Type	Field Name	Physical Table	Remote...
Alt	Attendance	Sem1_Scores.csv	Attendan...
Alt	Subject	Pivot	Pivot Fiel...
+	Score	Pivot	Pivot Fiel...
+	PHYSICS_CLEAN	Calculation	Calculati...
+	Attendance_clean	Calculation	Calculati...

Sem1_Scores.csv	Sem1_Scores.csv	Sem1_Scores.csv	Sem1_Scores.csv	Sem1_Scores.csv	Pivot	Score	Calculation	PHYSICS_CLEAN	Calculation	Attendance_clean
Student ID	Full Name	Dept info	Attendance	Subject						
S101	Aarav Sharma	CSE-01	85 %	Chemistry		88.0000		null		85.00
S101	Aarav Sharma	CSE-01	85 %	Math		85.0000		null		85.00
S101	Aarav Sharma	CSE-01	85 %	Physics		0.9000		null		85.00
S101	Aarav Sharma	CSE-01	85 %	Programming		95.0000		null		85.00
S102	priya VERMA	ECE-02	90 %	Chemistry		80.0000		null		90.00
S102	priya VERMA	ECE-02	90 %	Math		null		null		90.00
S102	priya VERMA	ECE-02	90 %	Physics		0.7500		null		90.00
S102	priya VERMA	ECE-02	90 %	Programming		85.0000		null		90.00

30°C Partly sunny

Search

ENG IN 10:56 20-08-2026

MVLU COLLEGE

TABLEAU PRACTICAL NO. 9

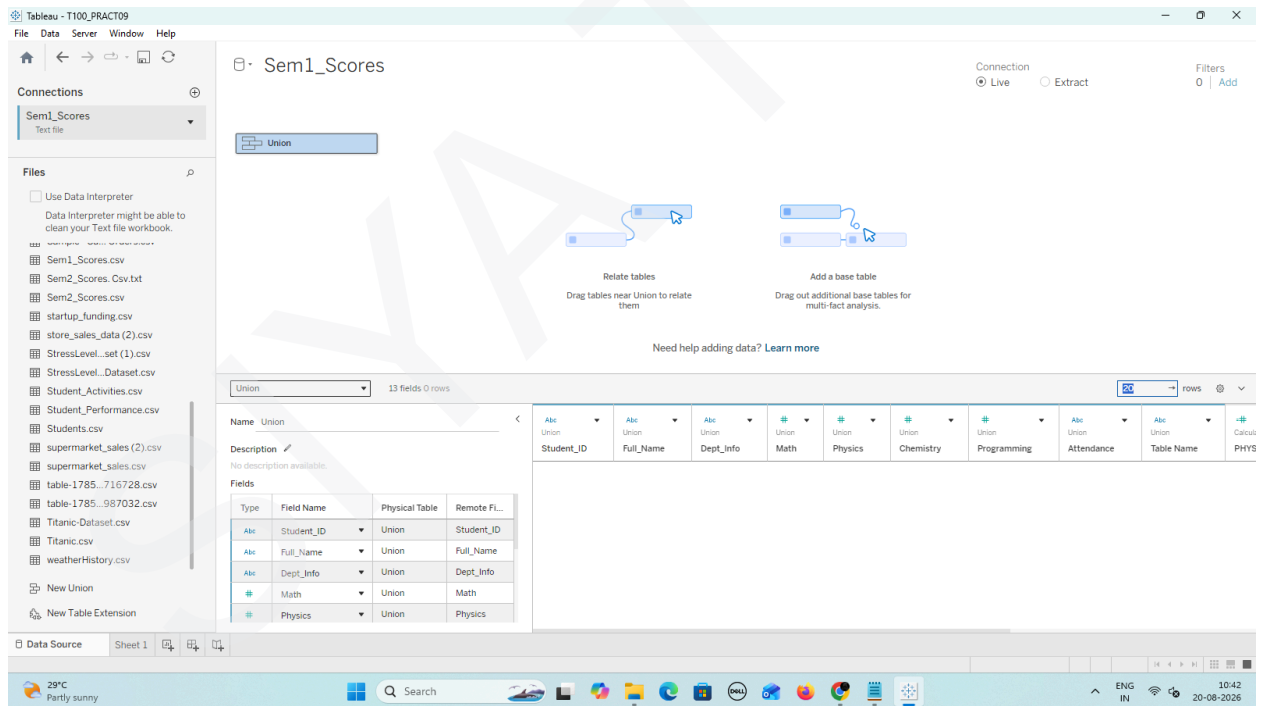
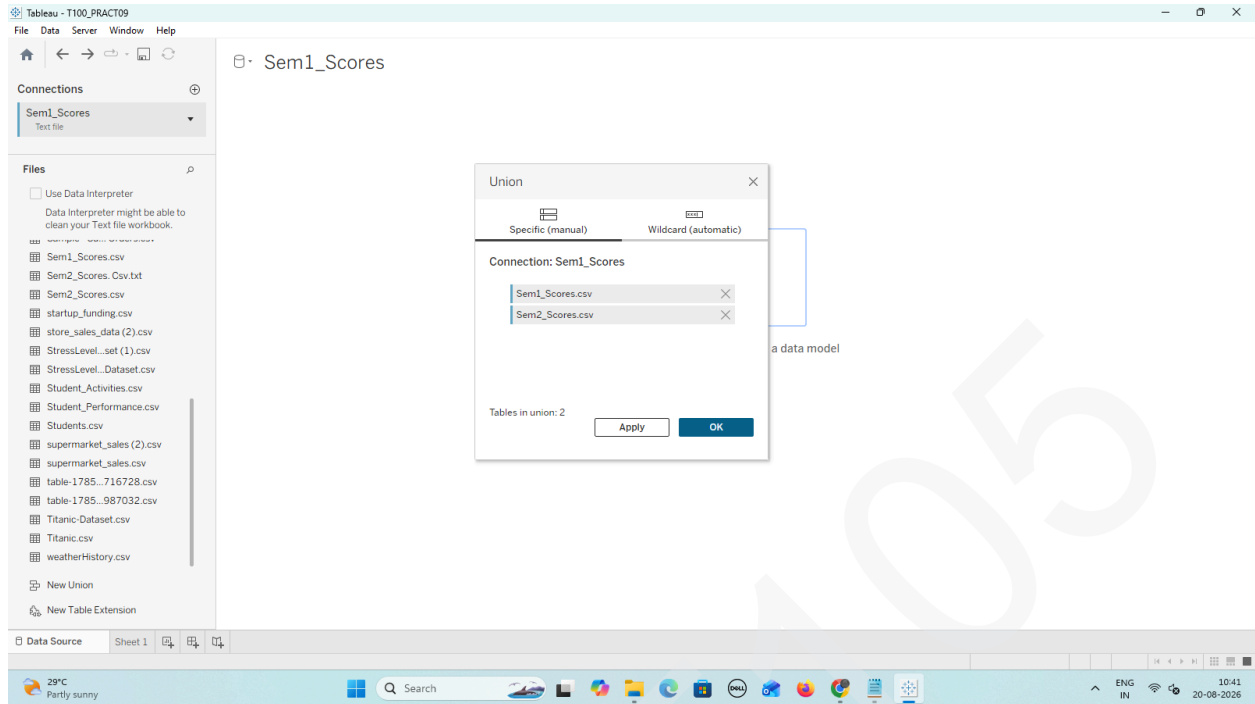
The screenshot shows the Tableau Desktop interface. A dialog box titled 'Clean_Score' is open, displaying the formula: `IFNULL(FLOAT(REPLACE(STR([Score]), ',' , '')), 0)`. The dialog has 'Apply' and 'OK' buttons. The background shows the 'Sem1_Scores' data source in the left pane and a blank worksheet in the center.

The screenshot shows the Tableau Desktop interface with the 'Sem1_Scores' data source selected. A context menu is open over the data source, showing options like 'Open...', 'Rename', 'Remove', etc. The 'Remove' option is highlighted. Below the menu, there are instructions to 'Relate tables' and 'Add a base table'. At the bottom, a table displays the data for 'Sem1_Scores.csv'.

Name	Sem1_Scores.csv	Student ID	Full Name	Dept info	Math	Physics	Chemistry	Programming	Attendance	PHYSICS_CLEAN
S103	Rohan Gupta	CSE-01	78	0.820000	null	88	78 %	0.820000		
S104	Sheha Patel	IT-03	92	0.880000	90	96	95 %	0.880000		
S105	aditya Rao	MECH-04	45	null	50	60	65 %	0.000000		
S106	Ananya Singh	CSE-01	88	0.910000	85	92	88 %	0.910000		
S107	Rahul Nair	ECE-02	null	0.700000	75	80	72 %	0.700000		
S108	Kavya Reddy	IT-03	95	0.940000	92	98	98 %	0.940000		
S109	Vikram Joshi	CSE-01	67	0.600000	58	72	80 %	0.600000		
S110	Pooja Iyer	ECE-02	82	0.850000	80	89	91 %	0.850000		

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TABLEAU PRACTICAL NO. 9



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TABLEAU PRACTICAL NO. 9

The screenshot shows the Tableau Desktop interface with a calculated field dialog box open. The dialog box contains the following code:

```
IF CONTAINS([Table Name], "Sem1")
THEN "Semester 1"
ELSEIF CONTAINS([Table Name], "Sem2")
THEN "Semester 2" ELSE "Other"
END
```

The dialog box also includes a message "The calculation is valid." and buttons for "Apply" and "OK". The background shows the Tableau workspace with a blank sheet and various toolbars.

The screenshot shows the Tableau Desktop interface with the 'Sem1_Scores' data source selected. The calculated field dialog box is open, showing the same code as in the previous screenshot. The background shows the Tableau workspace with a blank sheet and various toolbars.

MVLU COLLEGE

TABLEAU PRACTICAL NO. 9

Tableau - T100_PRACT09

File Data Server Window Help

Connections: Sem1_Scores (Text file)

Files: Use Data Interpreter, NetFix.csv, Order_Details.csv, Order_Detail_ctical_3.csv, Orders (1).csv, Retail_sales...all_sales.csv, Retail_sales.csv, Sample - Su... Orders.csv, Sem1_Scores.csv, Sem2_Scores.Csv.txt, Sem2_Scores.csv, startup_funding.csv, store_sales_data (2).csv, StressLevel_set (1).csv, StressLevel_Dataset.csv, Student_Activities.csv, Student_Performance.csv, Students.csv, New Union, New Table Extension

Sem1_Scores

Union

Department_Master.csv

Student_Activities.csv

Connection: Live (selected), Extract

Filters: 0 | Add

Union — Student_A...

10 rows

Name	Student_Activities.csv	Student_Activities.csv	Student_Activities.csv	Student_Activities.csv
	Student ID (Student Acti	Club Membership	Projects Completed	Certifications
Union	Operator	Student_Activities...		
Abc Student_ID	=	Abc Student ID (Stu		
Add more fields				
Performance Options				
Remove Relationship				
S101	Coding Club	3	2	
S102	Robotics Club	2	1	
S103	Coding Club	2	1	
S104	Debate Society	4	3	
S105	Sports Club	1	0	
S106	Coding Club	3	2	
S107	Robotics Club	2	1	
S108	Debate Society	4	3	

Data Source Sheet 1

30°C Partly sunny

Search

ENG IN 11:07 20-08-2026

Tableau - T100_PRACT09

File Data Worksheet Dashboard Story Analysis Map Format Server Window Help

Pages: Columns, Rows

Filters

Search

Tables: Department_Master..., Student_Activities.csv (Club Membership, Student ID (Student...), Certifications, Projects Completed, Student_Activities c...), Union (Attendance, Dept_Info, Full_Name, Semester, Student_ID, Table Name, Attendance_clean, Chemistry, Chemistry_Clean, Math, Math_clean, Physics, PHYSICS_CLEAN, Programming, Union (Count), Measure Names, Clean_Score, Measure Values)

Abc Measure Names

+ Clean_Score

Measure Values

Drop field here

Drop field here

Drop field here

Add data to visualize

Double-click or drag fields from the data pane.

Data Source Sheet 1

30°C Partly sunny

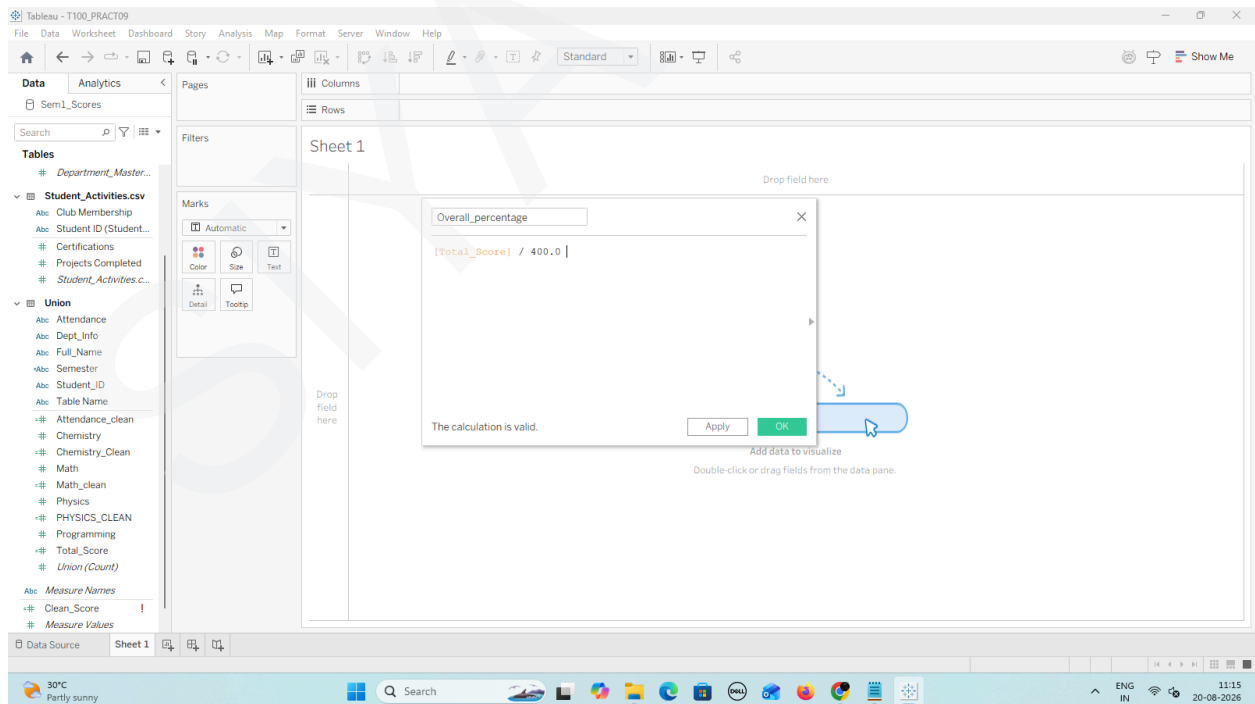
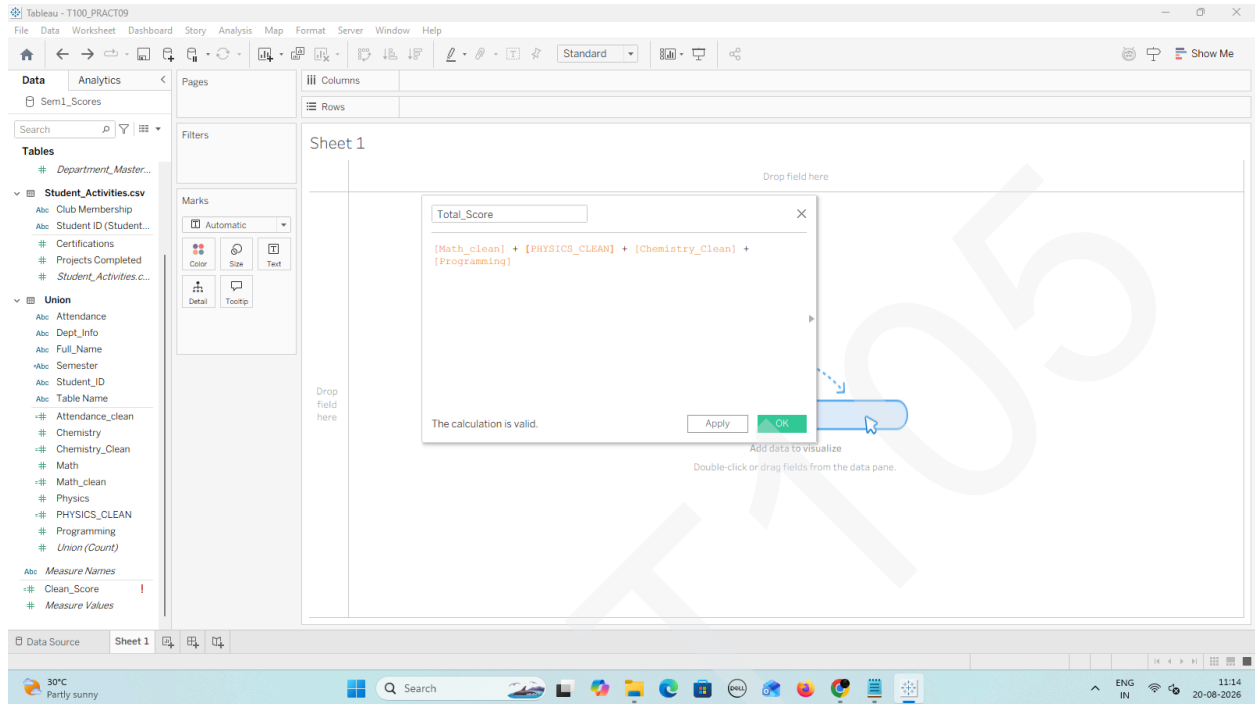
Search

ENG IN 11:10 20-08-2026

MVLU COLLEGE

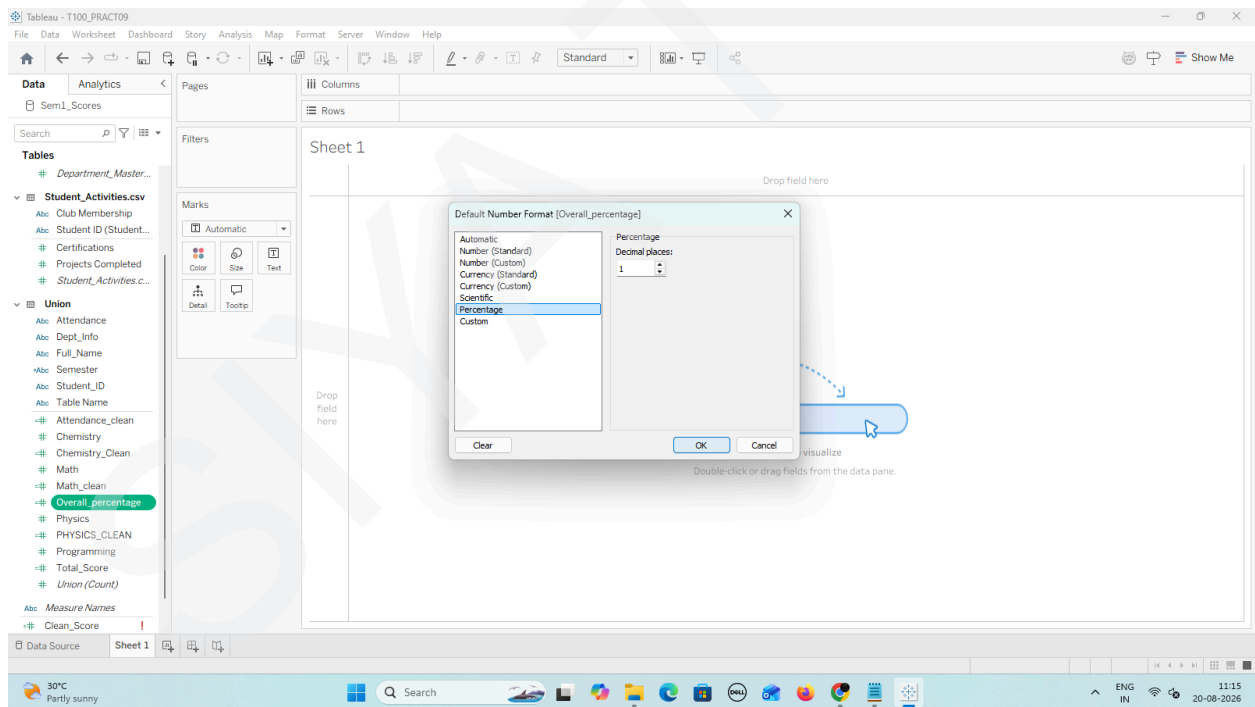
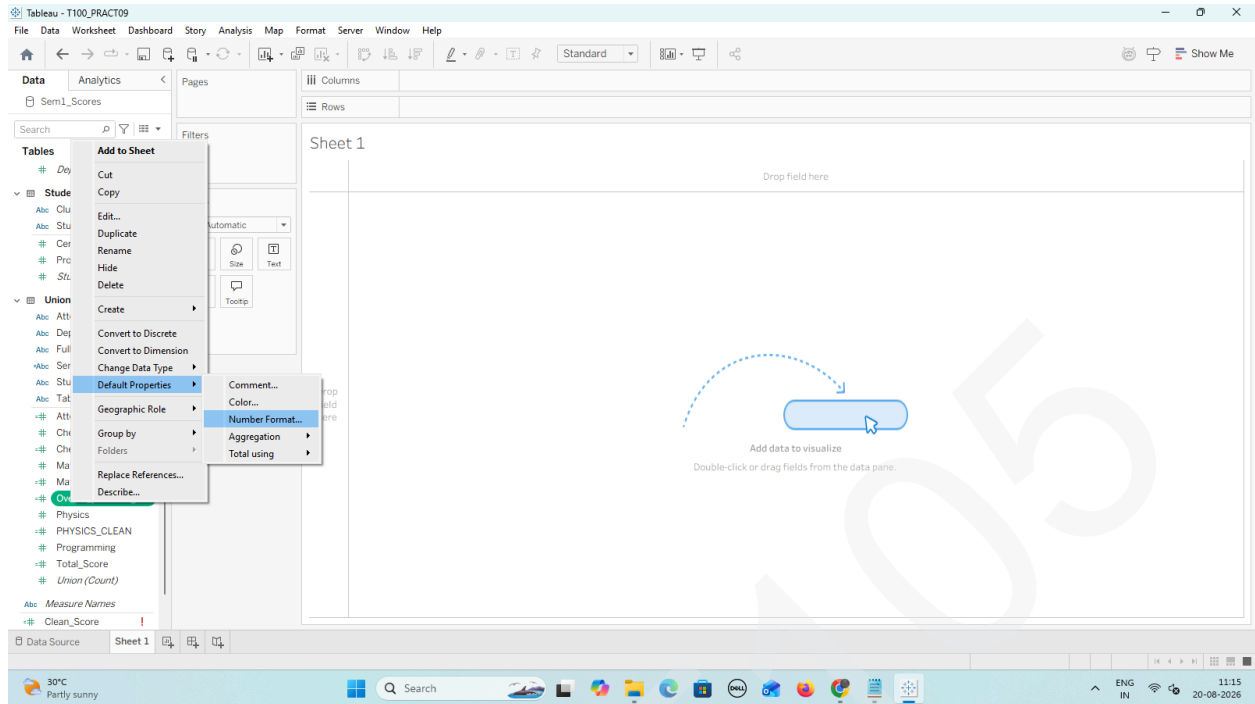
TABLEAU PRACTICAL NO. 9

- e) Restructure data for better visualization
- f) Handle different levels of detail in datasets



MVLU COLLEGE

TABLEAU PRACTICAL NO. 9



MVLU COLLEGE

TABLEAU PRACTICAL NO. 9

This screenshot shows the Tableau interface with a calculated field dialog box open. The dialog is titled "Performance_Band" and contains the following logic:

```
IF [Overall_percentage] >= 0.85
THEN "Distinction"
ELSEIF [Overall_percentage] >= 0.75
THEN "First Class" ELSEIF [Overall_percentage] >= 0.60
THEN "Second Class"
ELSE "Needs Academic Support"
END
```

The dialog also indicates "The calculation is valid." and has "Apply" and "OK" buttons. The background shows the Tableau workspace with "Sheet 1" and a list of data sources on the left, including "Student_Activities.csv" and "Union".

This screenshot shows the Tableau interface with a calculated field dialog box open. The dialog is titled "Attendance_Status" and contains the following logic:

```
IF [Attendance_clean] >= 75
THEN "Eligible (Good Standing)"
ELSE "Shortage Warning (<75%)"
END
```

The dialog also indicates "The calculation is valid." and has "Apply" and "OK" buttons. The background shows the Tableau workspace with "Sheet 1" and a list of data sources on the left, including "Student_Activities.csv" and "Union".

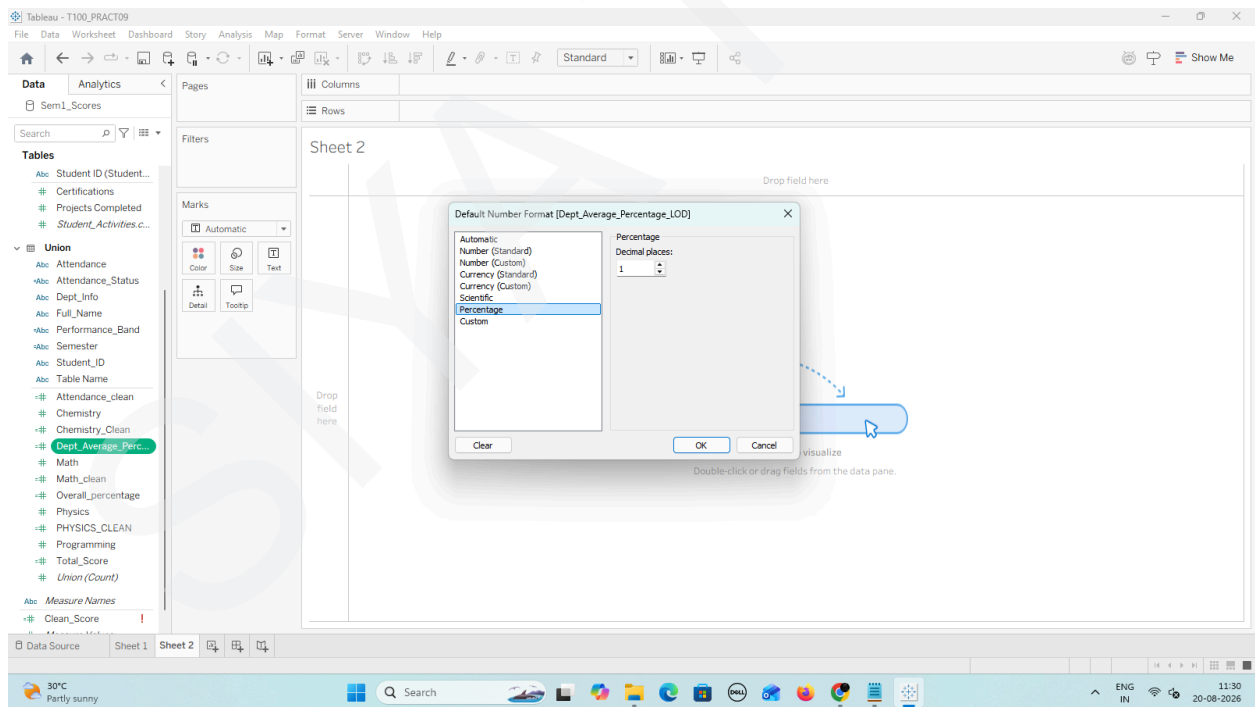
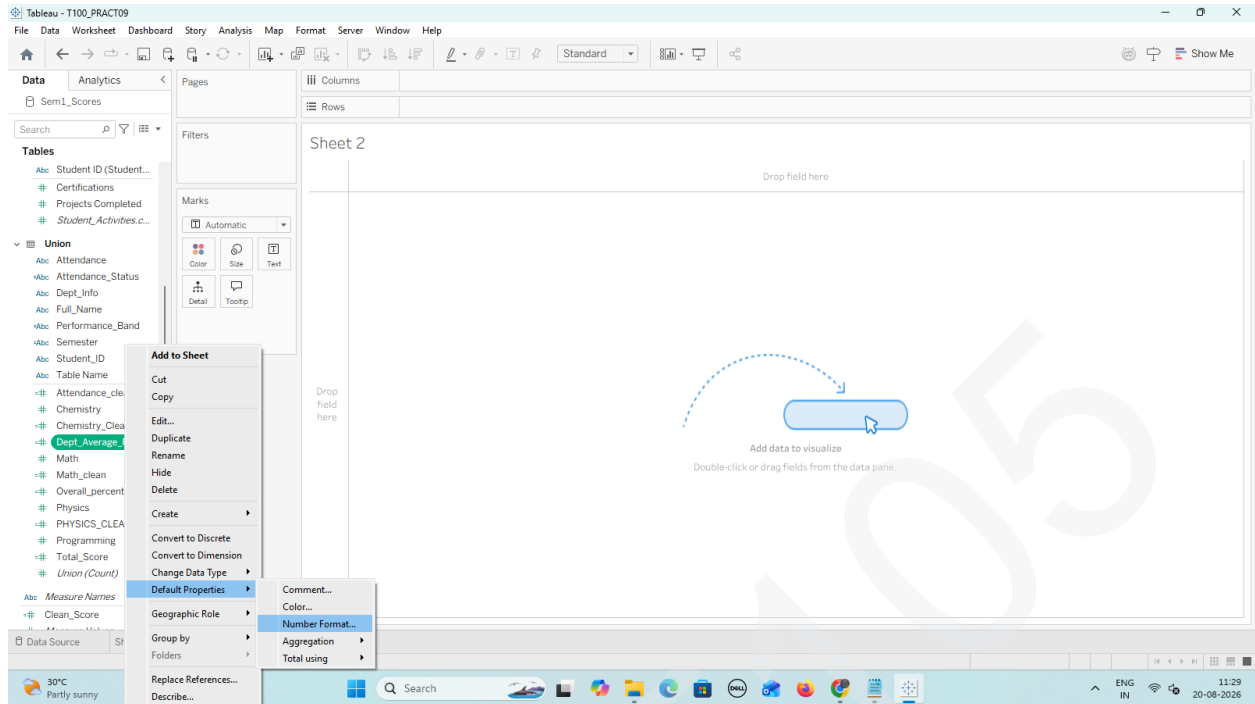
The screenshot displays the Tableau Desktop interface. The top menu bar includes File, Data, Worksheet, Dashboard, Story, Analysis, Map, Format, Server, Window, and Help. The main workspace shows a data table titled 'Sheet 1' with columns for Student ID, Full Name, Semester, and Marks. The table is filtered by Semester (Semester 1 and Semester 2) and Performance Band (Needs Academic Sup. and Second Class). The table contains data for two semesters across various subjects. The interface includes a sidebar with data sources, a top menu bar, and a bottom status bar.

	S101	S102	S103	S104	S105	S106	S107	S108	S109	S110		
Semester	Aarav Sharma	Priya Verma	priya VERMA	Rohan Gupta	Sneha Patel	aditya Rao	Aditya Rao	Ananya Singh	Rahul Nair	Kavya Reddy	Vikram Joshi	Pooja Iyer
Semester 1	67.2%	62.4%	41.4%	41.7%	69.7%	38.8%		66.5%	38.9%	71.5%	49.4%	63.0%
Semester 2	69.2%	62.4%	63.2%	72.0%		43.1%	68.5%	58.7%	72.7%	51.9%	65.5%	

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This screenshot shows the Tableau Desktop interface with a calculation editor open for the field 'Student_Overall_Career_Average'. The calculation is defined as $\{ \text{FIXED } [\text{Student_ID}] : \text{AVG}([\text{Overall_percentage}]) \}$. The interface includes a sidebar with data sources (Semi1_Scores), a list of tables (Union, Attendance, etc.), and a central workspace for visualizing data. The status bar at the bottom indicates the system is at 30°C and partly sunny.

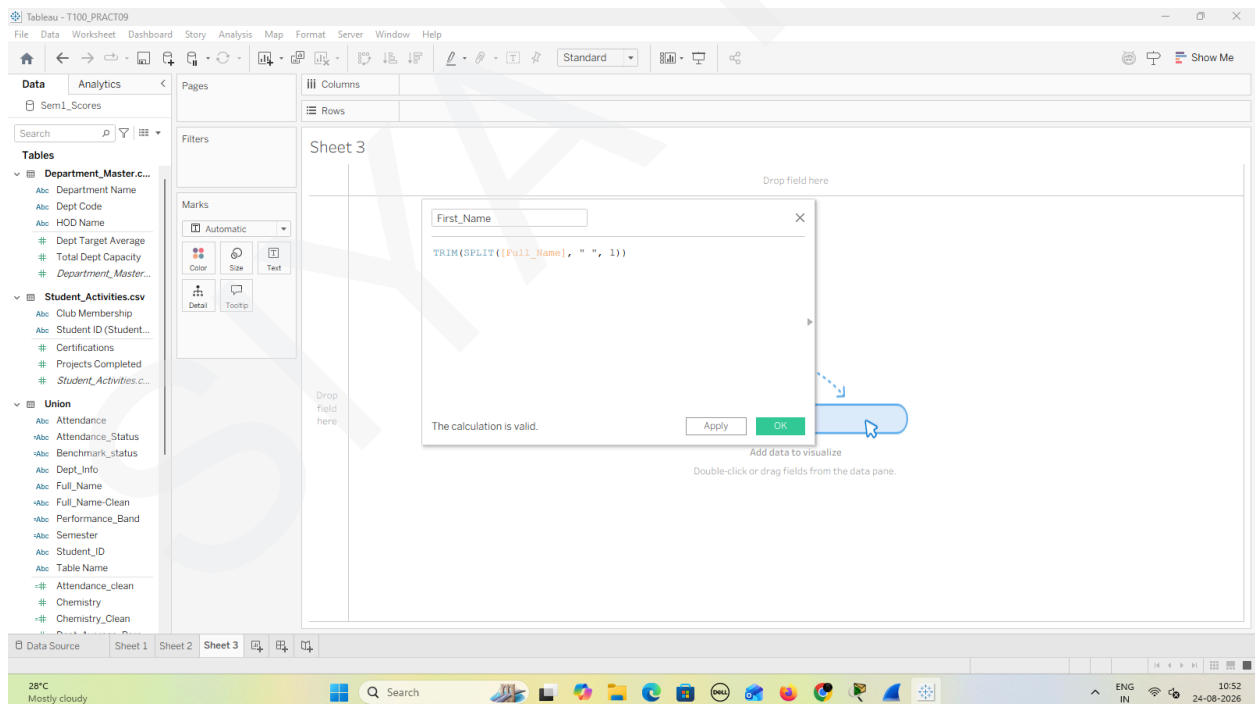
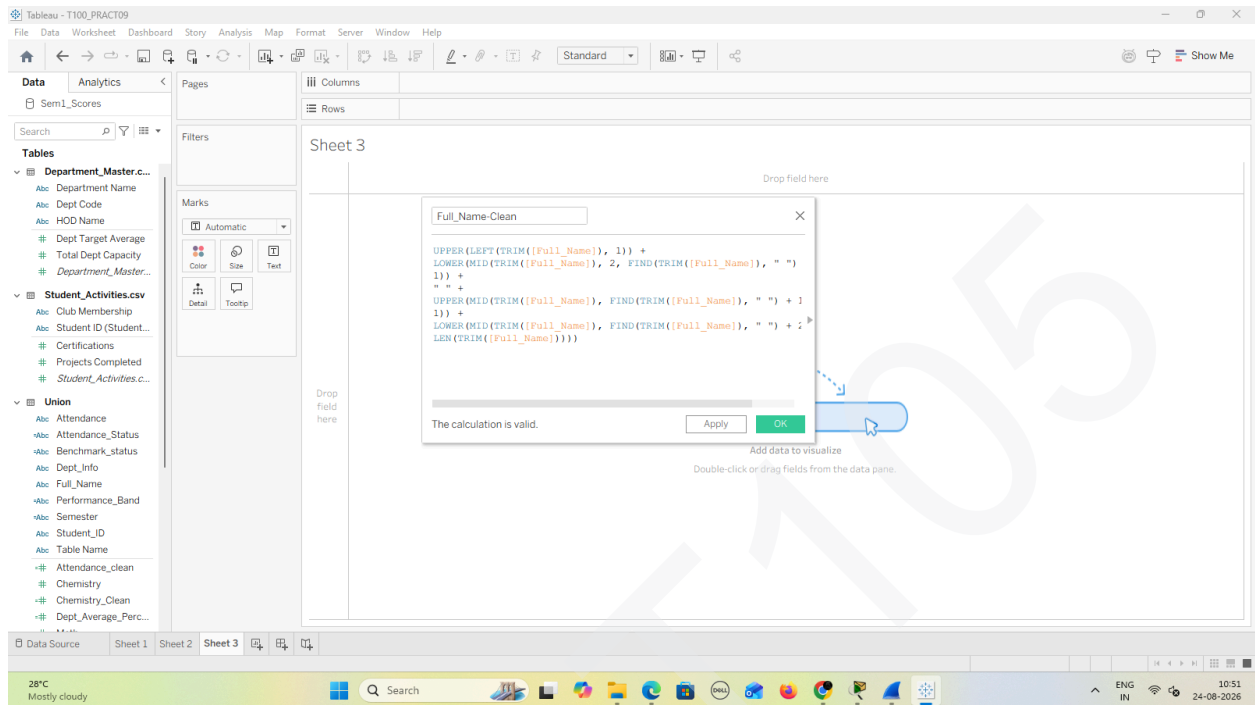
This screenshot shows the Tableau Desktop interface with a calculation editor open for the field 'Variance_From_Dept_Average'. The calculation is defined as $[\text{Overall_percentage}] - [\text{Dept_Average_Percentage_LOD}]$. The interface includes a sidebar with data sources (Semi1_Scores), a list of tables (Union, Attendance, etc.), and a central workspace for visualizing data. The status bar at the bottom indicates the system is at 30°C and partly sunny.

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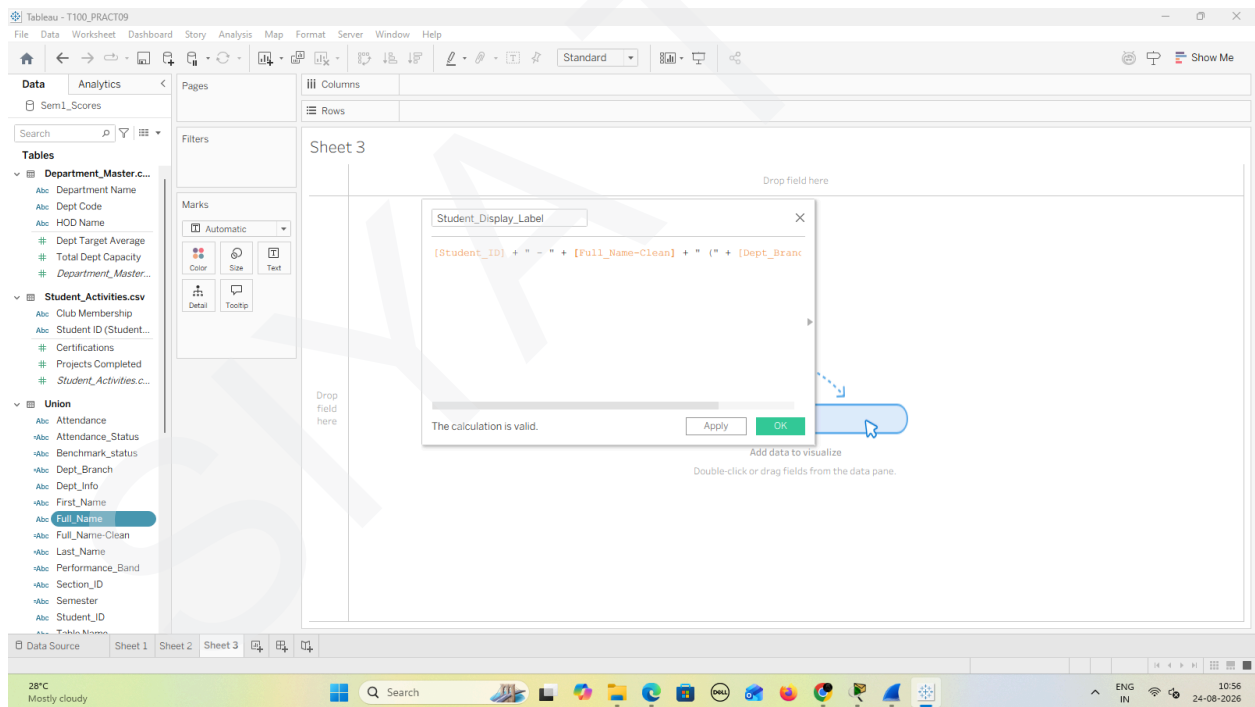
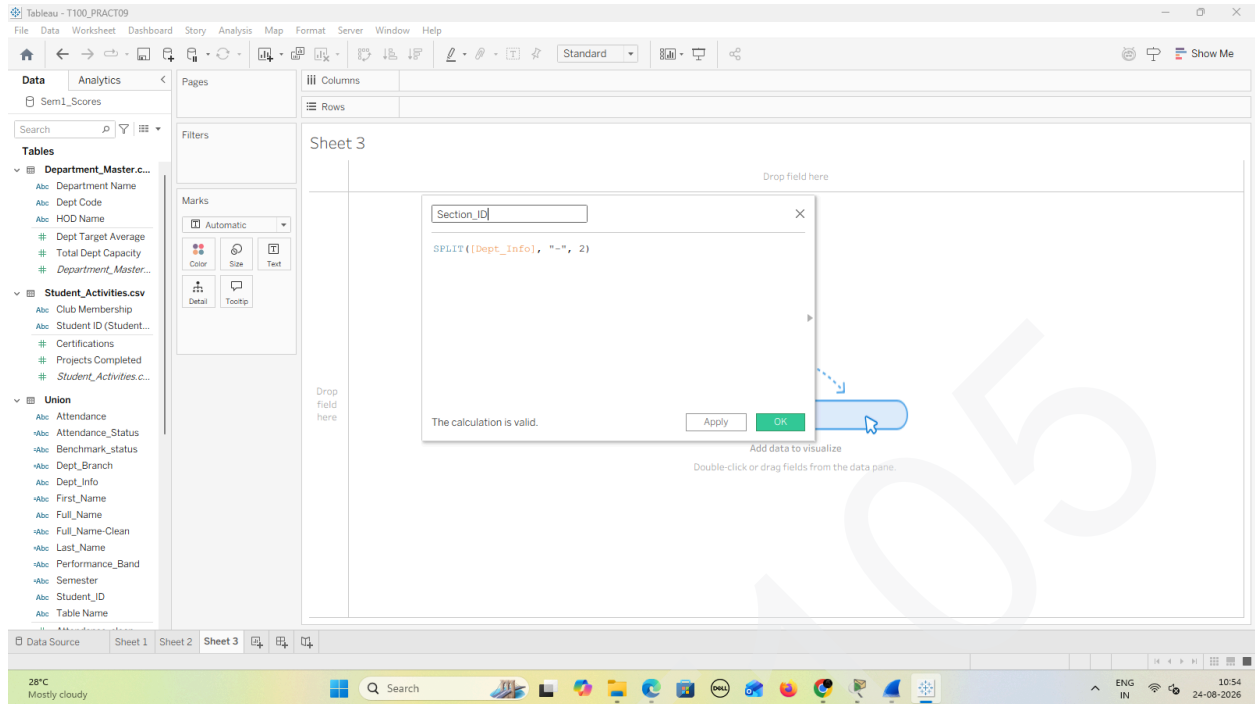
g) Apply advanced data cleaning techniques



The screenshot displays the Tableau Desktop interface. On the left, the 'Data' pane shows a list of tables and fields. The 'Department_Master.csv' table includes fields like 'Department Name', 'Dept Code', 'HOD Name', 'Dept Target Average', 'Total Dept Capacity', and 'Department_Master...'. The 'Student_Activities.csv' table includes 'Club Membership', 'Student ID (Student...', 'Certifications', 'Projects Completed', and 'Student_Activities.c...'. The 'Union' section lists fields from 'Attendance' and 'Attendance_Status' tables. The main view area shows 'Sheet 3' with a 'Drop field here' prompt. A dialog box is open in the center, titled 'Dept_Branch', containing the formula `SPLIT([Dept_Info], "-", 1)`. Below the formula, it states 'The calculation is valid.' and has 'Apply' and 'OK' buttons. A tooltip is visible over the 'OK' button, stating 'Add data to visualize' and 'Double-click or drag fields from the data pane.'

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